# Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI

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## Research Profile

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

## Education

**Ph.D., Electrical and Electronics Engineering** 2021 – ongoing
*Northfield Institute of Science and Technology*
**Thesis:** Lightweight deep networks for real-time enhancement of aerial imagery
**Supervisor:** Prof. R. Iyer (fictional)

**M.Tech., Avionics** 2010
*Coastal Institute of Aeronautical Engineering, Chennai*
**Grade:** CGPA 8.6/10
**Thesis:** Fault-tolerant data acquisition for airborne test instrumentation

**B.E., Electronics and Communication Engineering** 2004
*Northfield Institute of Science and Technology, Northfield University, Pune*
**Grade:** First Class with Distinction

## Publications

[1] **A. Sharma** and R. Iyer, "Knowledge-distilled lightweight networks for drone image super-resolution," accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, "Benchmarking embedded GPUs for real-time image enhancement," Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, "Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs," Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, "An IoT telemetry framework for upper-air ground stations," Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, "Edge telemetry dashboards for upper-air ground stations," Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, "A low-cost GNSS receiver chain for radiosonde tracking," Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

## Professional Experience

**Scientist E** Jul 2016 – Present
*Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi*
Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.

*Additional charge:* **Workshop In-charge**, Upper Air Division Workshop Apr 2019 – Present
Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

*Nodal officer:* **Nodal Officer, IoT and AI/ML Initiatives** Jan 2021 – Present

**Senior Design Engineer** Jun 2004 – Jun 2016
*Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru*
Integration and qualification testing of airborne electronics and UAV payloads.



*Fictional sample profile — not a real person*

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.

**Graduate Engineer** Aug 2002 – May 2004
*Test Systems Group, Meridian Electronics Limited, Bengaluru*
Built automated test equipment for avionics line-replaceable units.
- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

## Research Projects

**Edge super-resolution pipeline for UAV imagery** *(Ongoing)* Sep 2021 – Present
*Principal investigator, Kestrel Atmospheric Instrumentation Laboratory*
Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.
- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.

## Research Grants & Funding

**Seed grant: edge-AI enhancement of aerial imagery** *(Completed)* Apr 2023 – Mar 2025
*Principal investigator, Example Science Foundation (fictional)*
Two-year seed project on lightweight super-resolution networks for embedded GPUs.
**Reference:** ESF/SEED/2023/017

## Talks & Presentations

**Invited talk: Edge AI for atmospheric observation** 2024
*Example Symposium on Applied AI*

## Awards & Honours

**Best Paper Award** 2023
*Example Conference on Instrumentation (ECI 2023)*

## Professional Service

**Reviewer, Example Journal of Atmospheric Instrumentation** 2022 – Present
*Example Journal of Atmospheric Instrumentation*

**Co-supervisor, two M.Tech. dissertations in embedded vision** 2024 – 2025
*Example Institute of Technology (fictional)*

## Professional Memberships

**Member, Institution of Example Engineers** 2010 – Present
*Institution of Example Engineers*

## Certifications & Training

**Deep Learning for Edge Devices** Mar 2022
*Example Academy of Applied Computing*
**Duration:** 40 hours
**Credential:** EAAC-DL-0042
**Valid until:** Mar 2029

## Technical Skills

**Avionics:** MIL-STD-1553 / ARINC 429, Avionics integration
**Scientific instrumentation:** Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems
**Embedded systems:** Embedded C, ESP32 / STM32 microcontrollers
**Edge AI:** NVIDIA Jetson deployment

## Languages

English, Hindi


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